



UNIVERSITY OF
CENTRAL FLORIDA

EEL 4742C - Embedded Systems

Section: 0001

*College of Engineering and Computer
Science*

Department of Electrical and
Computer Engineering

Course Information

Term: Fall 2026

Class Meeting Days: TR

Class Meeting Time: 09:00AM - 10:15AM

Class Meeting Location: BA1 O116

Modality: P

Credit Hours: 3.00

Instructor Information

Name: Hadi Kamali

Title: Assistant Professor

Office Location: HEC 435 (or Zoom by prior Email)

Office Hours

Tuesdays/Thursdays 1:30PM - 3:00PM

Email: kamali@ucf.edu

Office Hours

Tuesdays/Thursdays 1:30PM - 3:00PM @ HEC 435 (or Zoom by prior email)

Teaching Assistants

The teaching assistant (TA) for this course will be responsible for assisting with conducting lab sessions and Grading.

This information might be updated in the first weeks of the semester:

TA (grader for Section 0001): Sk Samiul Reza (email:

sksamiul.reza@ucf.edu)

Course Description

EEL 4742C ENGR-ECE 3(3,2) Embedded Systems: PR: Earn a minimum grade of C in each of the following: EEL3801C - Computer Organization (4) Microcontroller systems, assembly language programming, data representation, memory and device interfacing, timing analysis, parallel and serial communication, timers, interrupts, signal interfacing considerations, and applications. Fall, Spr, Sum

The following topics will be covered in this class:

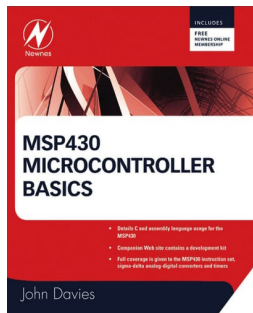
- Introduction to Embedded Systems
- MSP430 Microcontrollers
- Masking Arithmetic
- Flashing LEDs and Reading Push Buttons
- Using the Timer
- Interrupts
- Low-power Modes
- LCD Display
- Advanced Timer Features (multiple channels, timer output)
- Concurrent Event Handling with Interrupts
- UART Communication
- I2C Communication
- ADC Converter (SAR Type)
- SPI Communication
- Pixel Graphics
- May be discussed if time permits: Clock Modules, MSP430 Architecture, Real-time Operating System (RTOS)

Student Learning Outcomes

After successful completion of this course, students will be able to:

- understand the embedded systems basics (chip organization, programming environment);
- be able to do masking arithmetics (bit manipulations);
- understand the timer module's various modes and be able to program it;
- understand interrupts and be able to program them;
- understand the low-power modes and be able to use them in codes;
- be able to understand the basics of the segmented LCD display and program it;
- understand the push button debouncing algorithms and be able to program them;
- understand the communication protocols SPI, I2C and UART;
- understand the basics of the analog-to-digital converter (SAR type);
- be able to understand the basics of the pixel display and program it.

Required Course Materials and Resources



MSP430 Microcontroller Basics

ISBN: 9780750682763

Authors: Davies, John H.

Publisher: Elsevier S & T

Publication Date: 2008-08

Edition: 1

Online Access: <https://go.openathens.net/redirector/ucf.edu?url=https%3A%2F%2Fwww.sciencedirect.com%2Fbook%2F9780750682763>

MSP430FR6989 Chip Data Sheet (slas789d)

Online Access: <https://www.ti.com/lit/ds/slas789d/slas789d.pdf>

LaunchPad Board User's Guide (slau627a)

Online Access: <https://www.ti.com/lit/ug/slau627a/slau627a.pdf>

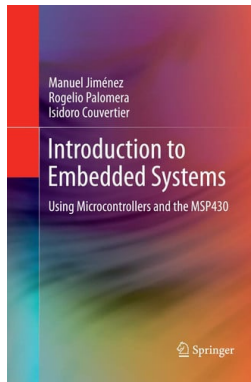
Booster Pack User's Guide (slau599b)

Online Access: <https://www.ti.com/lit/ug/slau599b/slau599b.pdf>

MSP430FR6xx Family User's Guide (slau367o)

Online Access: <https://www.ti.com/lit/ug/slau367p/slau367p.pdf>

Recommended Course Materials



**Introduction to Embedded Systems Using
Microcontrollers and the MSP430**

ISBN: 9781493944286

**Authors: Manuel Jiménez, Rogelio Palomera, Isidoro
Couvertier**

Publisher: Springer New York

Publication Date: 2017-04-30

Edition: 1

Course Assessment and Grading Procedure

- Homework 15%
- Lab/Project: 20%
- Midterm 1: 20%
- Midterm 2: 20%
- Final Exam: 25%
- Quizzes (Bonus): 10%

Grading Scale

Grading Scale	
Letter Grade	Percentage
A	90-100%
B	80-90%
C	70-80%
D	60-70%
F	0-60%

Communication Expectations

The preferred method of communication for this course is through

Webcourses/Canvas Inbox or university email. Students should use their UCF email account and include EEL 4742 in the subject line when contacting the instructor. I will normally respond to course-related messages within 24–48 hours.

Course announcements, assignment updates, schedule changes, and other important information will be posted through Webcourses/Canvas Announcements. Students are responsible for checking Webcourses and their UCF email regularly.

For questions that require detailed discussion or assistance with homework, students are encouraged to attend office hours or schedule an appointment.

Policies for Course Grade

General Policies:

- **Class announcements are made on Webcourses.**
- **Lab attendance is required. If you need to miss a lab session, inform the instructor/TA beforehand.**
- **You will have lab access for after hours so you can work on the lab beyond the lab session.**
- **Ensure that uploads to Webcourses are successful. After uploading a document, download it back from the server and open it on your computer to ensure that the server has received the file.**
- **If you need to request a grade revision, please do so within one week from when the grades are posted so that the TAs don't get overloaded with requests at the end of the semester.**
- **Late assignments may or may not be accepted depending on how late they are. Consult with the instructor.**
- **Make-up exams require a university accepted and documented reason. Consult with the instructor.**

Lab Policies:

- **Each lab's work needs to be checked off with the TA in person. During the checkoff, the TA will ask questions about your work. Your report will not be graded if you didn't check it off with the TA.**
- **In your report, include an introduction to the lab and a conclusion that summarizes the main learning points. For each part in the lab, write a paragraph that explains the main points of your work. Do not copy sentences verbatim from the lab manual.**
- **Code must be copy/pasted as text (not as picture) in the report.**

- Lab reports are due on the day of your lab before your lab start time. E.g., if your lab is on Monday at 2pm, you should submit the previous week's lab report by Monday at 2pm.
- Each lab is graded based on 20 total points.

Make-up Assignments for Authorized University Event or Co-curricular Activities

Students who represent the university at an authorized event or activity (for example, student athletes) and who are unable to meet a course deadline due to a conflict with that event must provide the instructor with documentation in advance to arrange a make-up. No penalty will be applied if the student gives advance notice and communicates with the instructor following UCF policy. In the case of an authorized university activity, it is your responsibility to show the instructor a signed copy of the Program Verification Form for which you will be absent, prior to the class in which the absence occurs or due date you need extended. For more information, see [UCF Policy 4-401](#).

Make-up Assignments for Religious Observances

A student who desires to observe a religious holy day of his or her religious faith must notify their instructor as soon as practicable prior to the observance. Students who are absent because of religious observances will be permitted a reasonable amount of time to complete any missed work. For more information, see [UCF Regulation 5-020](#).

Artificial Intelligence (AI) Use Policy

Use of Generative Artificial Intelligence (GenAI) is prohibited. Use of GenAI tools via website, app, or any other access, is not permitted in this class. All components of assignments in this course must be independently and originally completed by the student. Representing work created by GenAI as your own work will be treated as plagiarism.

Disability Access & Accommodations

The University of Central Florida is committed to providing equal access to all students with disabilities (ADHD, learning disabilities, Autism, chronic medical conditions, physical disabilities, etc.). To receive consideration for reasonable disability-related course accommodations, disabled students must contact Student Accessibility Services (SAS) and complete the steps required for SAS to review accommodation requests. More information can be found on the UCF [Student Accessibility Services](#) website under the Start Here tab or by contacting SAS directly (Ferrell Commons 185; sas@ucf.edu; Phone - 407-823-2371).

Approved accommodations are shared with course instructors via the SAS Course Accessibility Letter. Implementing certain accommodations may require discussion about specific considerations of the course design, course learning objectives, and the individual academic and course challenges experienced by the student. While students with disabilities or chronic health needs are also encouraged to discuss any course concerns with professors in addition to contacting SAS, professors are not required to facilitate disability-related adjustments to the course unless the professor has received a Course Accessibility Letter from SAS that outlines approved accommodations.

Academic Integrity

Students should familiarize themselves with UCF's Code of Conduct at Student Conduct and Integrity Office. According to Section 1, "Academic Misconduct," students are prohibited from engaging in:

1. **Academic misconduct is defined as any submitted work or behavior that obstructs the instructor of record's ability to accurately assess the student's understanding or completion of course materials or degree requirements (e.g., assignment, quiz, and/or exam). Examples of academic misconduct include but are not limited to: plagiarism, unauthorized assistance to complete an academic exercise; unauthorized communication with others during an examination, course assessment, or project; falsifying or misrepresenting academic work; providing misleading information**

- to create a personal advantage to complete course/degree requirements; or multiple submission(s) of academic work without permission of the instructor of record.
2. Any student who knowingly helps another violate academic behavior standards is also in violation of the standards.
 3. Commercial Use of Academic Material. Selling of course material to another person and/or uploading course material to a third-party vendor without authorization or without the express permission of the University and the instructor of record. Course materials include but are not limited to class notes, the instructor of record's slide deck, tests, quizzes, labs, instruction sheets, homework, study guides, and handouts.
 4. Soliciting assistance with academic coursework and/or degree requirements. The solicitation of assistance with an assignment, lab, quiz, test, paper, etc., without authorization of the instructor of record or designee is prohibited. This includes but is not limited to asking for answers to a quiz, trading answers, or offering to pay another to complete an assignment. It is considered Academic Misconduct to solicit assistance with academic coursework and/or degree requirements, even if the solicitation did not yield actual assistance (for example, if there was no response to the solicitation).

Responses to Academic Dishonesty, Plagiarism, or Cheating

Students should also familiarize themselves with the procedures for academic misconduct in UCF's student handbook, [*The Golden Rule*](#). UCF faculty members have a responsibility for students' education and the value of a UCF degree, and so seek to prevent unethical behavior and respond to academic misconduct when necessary. Penalties for violating rules, policies, and instructions within this course can range from a zero on the exercise to an "F" letter grade in the course. In addition, an Academic Misconduct report could be filed with the Office of Student Conduct and Academic Integrity, which could lead to disciplinary warning, disciplinary probation, or deferred suspension or separation from the University through suspension, dismissal, or expulsion with the addition of a "Z" designated on one's transcript.

Being found in violation of academic conduct standards could result in a student having to disclose such behavior on a graduate school application, being removed from a leadership position within a student organization, the recipient of scholarships, participation in University activities such as

study abroad, internships, etc.

Let's avoid all of this by demonstrating values of honesty, trust, and integrity. No grade is worth compromising your integrity and moving your moral compass. Stay true to doing the right thing: take the zero, not a shortcut.

Title IX

Title IX prohibits sex discrimination, including sexual misconduct, sexual violence, sexual harassment, and retaliation. If you or someone you know has been harassed or assaulted, you can find resources available to support the victim, including confidential resources and information concerning reporting options at [Let's Be Clear](#) and [UCF Cares](#).

For more information on access and community engagement, Title IX, accessibility, or UCF's complaint processes contact:

- Title IX – ONAC – [Office of Nondiscrimination & Accommodations Compliance](#) & askanadvocate@ucf.edu
- Disability Accommodation – Student Accessibility Services – [Student Accessibility Services](#) & sas@ucf.edu
- [Access and Community Engagement](#) (including the Ginsberg Center for Inclusion and Community Engagement, Military and Veteran Student Success, and HSI Initiatives)
- UCF Compliance and Ethics Office – [Compliance, Ethics, and Risk Office](#) & complianceandethics@ucf.edu
- The [Ombuds Office](#) is a safe place to discuss concerns.

Reporting an Incident or Issue

If you believe you have experienced discrimination by any faculty or staff member, contact the Office of Nondiscrimination & Accommodations Compliance via the [ONAC website](#) or at 407-823-1336. You can also choose to report using the UCF Integrity Line either anonymously or as yourself at 1-855-877-6049 or by using the [online form](#). UCF cares about you and takes every report seriously. For more information see the [Reporting an Incident or Issue Webpage](#).

Deployed Active-Duty Military Students

Students who are deployed active-duty military and/or National Guard personnel and require accommodation should contact their instructors as soon as possible after the semester begins and/or after they receive notification of deployment to make arrangements.

Campus Safety

At UCF's Public Safety and Police, safety is the top priority. Emergencies on campus are rare, but if one should arise, it's important to be familiar with some basic safety and security concepts.

- In an emergency, always dial 911.
- Every UCF Classroom has an Emergency Procedure Guide posted on a wall near the door, which will show you how to respond to a variety of situations. This guide can also be found online [here](#).
- In the event of an active threat, remember **AVOID, DENY, DEFEND**. Choose the best course of action and act immediately. Watch the video [here](#) to learn more.
 - **AVOID.** Pay attention to your surroundings and have an exit plan. Get as much distance and as many barriers between you and the threat as quickly as possible.
 - **DENY.** When avoiding is difficult or impossible, deny the threat access to you and your space. Lockdown by creating barriers, turning the lights off and remaining quiet and out of sight. Make sure your phone is silenced, but do not turn it off.
 - **DEFEND.** When you are unable to put distance between yourself and the threat, be prepared to protect yourself. Commit to your actions, be aggressive and do not fight fairly. Do whatever it takes to survive.
- For emergencies on campus, UCF will utilize the [UCF Alert](#) system. All UCF students, faculty, and staff are automatically enrolled to receive these email and text alerts, however, it's a good idea to frequently ensure your [contact information is up to date](#).

Financial Aid Accountability

All instructors are required to document students' academic activity at the beginning of each course. In order to document that you began this course, please complete this activity by the end of the first week of classes or as

soon as possible after adding the course. Failure to do so may result in a delay in the disbursement of your financial aid.

Class Schedule

This is tentative. Please refer to the "schedule" page for the latest updates.

Course Schedule		
Week	Topic	Assigned Reading(s)
1 (8/25 & 8/27)	Intro to Embedded Systems & Programming	No Lab
2 (9/1 & 9/3)	Microcontrollers (MSP430, Interfaces, & Peripherals)	Lab 1
3 (9/8 & 9/10)	Embedded System Designs - Timer	Homework 1 assignment
4 (9/15 & 9/17)	FSM and Interrupt	Lab 2 Homework 1 due
5 (9/22 & 9/24)	Low Power Modes and LCD	Lab 3 Homework 2 assignment
6 (9/29 & 10/1)	UART	Lab 4 Homework 2 due Midterm 1
7 (10/6 & 10/8)	UART	Lab 5
8 (10/13 & 10/15)	I2C	Lab 6 Homework 3 assignment
9 (10/20 & 10/22)	ADC	Lab 7 Homework 3 due
10 (10/27 & 10/29)	ADC & SPI	Lab 8 Homework 4 assignment
11 (11/3 & 11/5)	SPI & Timer Channels	Homework 4 due Midterm 2
12 (11/10 & 11/12)	Timer I/O	
13 (11/17 & 11/19)	Advanced Interrupts (Concurrent)	Project due Homework 5 assignment
14 (11/24 & 11/26)	Thanksgiving - No Class	
15 (12/1 & 12/3)	RTOS & Final Exam Review	Homework 5 due

Week	Topic	Assigned Reading(s)
16 (12/8 & 12/10)		Final Exam Thursday, December 10, 2026 7:00 AM – 9:50 AM